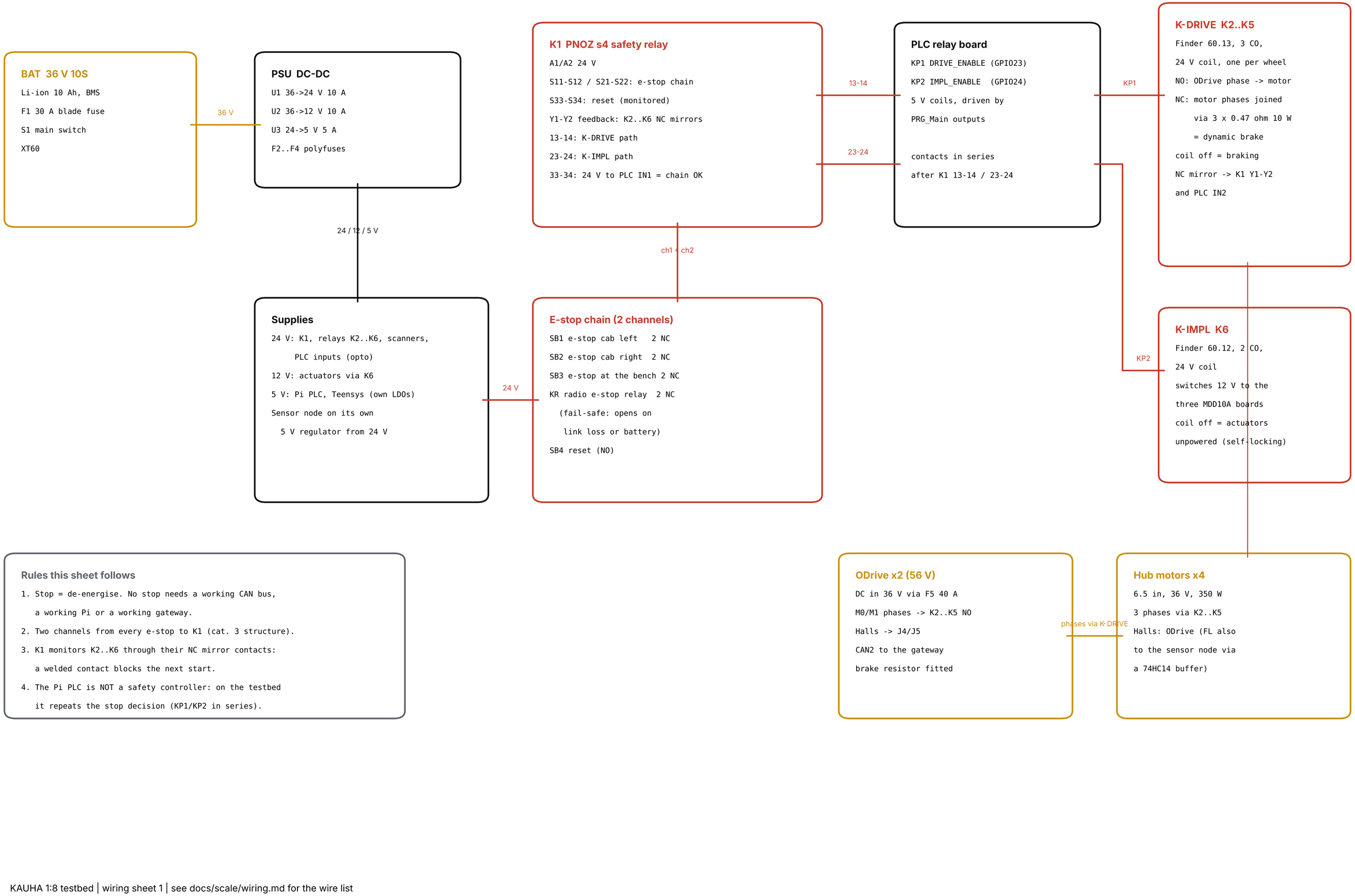
KAUHA 1:8 testbed - sheet 1: power, stop chain and hardwired outputs

Red: safety chain (24 V). Amber: 36 V traction. Black: logic. Every stop path de-energises: a broken wire, a dropped relay or a dead PLC all stop the machine.



Rules this sheet follows

1. Stop = de-energise. No stop needs a working CAN bus,
   a working Pi or a working gateway.
2. Two channels from every e-stop to K1 (cat. 3 structure).
3. K1 monitors K2..K6 through their NC mirror contacts:
   a welded contact blocks the next start.
4. The Pi PLC is NOT a safety controller: on the testbed
   it repeats the stop decision (KP1/KP2 in series).

KAUHA 1:8 testbed - sheet 2: CAN bus and signals

Machine bus: 500 kbit/s, 29-bit J1939 identifiers, hardware/can/kauha_machine.dbc. Two separate twisted pairs, each with 120 ohm at both ends. The gateway's segment guard is the only connection between them. Motor bus (CAN2) is separate.

Frame timing and timeouts (PLC side)

```
RemoteHeartbeat 20 Hz   timeout 0.5 s  -> REMOTE_LINK
DriveCmd        20 Hz   timeout 0.2 s  -> WATCHDOG
MachineState1   40 Hz   stale 0.1 s    -> SPEED_DISCREPANCY
SafeSensors     40 Hz   stale 0.1 s    -> SCANNER_FAULT
ScannerStatus   40 Hz   stale 0.1 s    -> SCANNER_FAULT
ZoneStatus      10 Hz   stale 0.5 s    -> POSITION_INVALID (2 s)
```

Every safety-relevant frame

byte 6 low nibble: alive counter (+1 per frame)

byte 7: CRC-8 SAE J1850 over data ID + bytes 0..6

data ID = PGN low byte (unique per message)

checked in FB_E2ERx (PLC) and kauha_e2e.c (firmware)

both tested bit for bit against kauha/can/e2e.py

open segment: + FrameAuth (AES-128 CMAC, 32 bit,
24-bit freshness), checked by the segment guard

Connectors

CAN: 4-pin M8 or JST-XH: CAN_H, CAN_L, GND, shield

Stub length < 0.3 m at 500 kbit/s

E-stop and relay wiring: 0.5 mm2, ferrules,
two colours for the two channels

Traction: 2.5 mm2 silicone, XT60/XT90



Safety PLC 0x81

Pi 4 + CAN HAT (can0)

CODESYS PRG_Main 25 ms

tx SafetyStatus 40 Hz

rx all frames below

GPIO: IN1..IN8, OUT1..OUT9

Sensor node 0x84

Teensy 4.0

AS5600 hinge (Wire)

AS5600 boom (Wire1)

FL Halls via 74HC14

tx SafeSensors 40 Hz

Field evaluator 0x85

evaluator profile only:

Pi + 2 x LD19 lidar

kauha/scale/field_eval.py

tx ScannerStatus 40 Hz

(rated scanners: wired to
PLC IN3..IN8 instead)

Gateway 0x83 + segment guard

Teensy 4.1: CAN1 safety, CAN3 open

copies safety -> open

open -> safety: 4 command frames,
only with a valid FrameAuth

tx MachineState1/2, AuthChallenge

CAN2 -> ODrives, PWM -> MDD10A

Autonomy 0x80

Pi 5 / Jetson, ROS 2

kauha_ros can_bridge (key_file)

tx DriveCmd, ImplementCmd
each followed by FrameAuth

rx SafetyStatus, MachineState

Remote 0x82 / zones 0x86

supervisor laptop:

tx RemoteHeartbeat 20 Hz

tx ZoneStatus 10 Hz
each followed by FrameAuth

(console + zone switches)